

PRIYA BALAKRISHNAN

ProfileWebsite  github.com/Bpriya42  linkedin.com/in/priya-balakrishnan  bpriya2001@gmail.com

EDUCATION

The University of Massachusetts Amherst

Amherst, MA

Master of Science in Computer Science - ML / AI concentration (GPA 3.8 / 4.0)

Anticipated - Sept 2024 - May 2026

- **Relevant Courses:** Statistics, Neural Networks, Systems for Data Science, Natural Language Processing, Computer Vision, Software Engineering, Optimization in Machine Learning.

The University of Queensland

Brisbane, Australia

Bachelor of Engineering (Hons) - Electrical and Computer Engineering

Jul 2018 - Jul 2023

- **Relevant Courses:** Data structures and Algorithms, Object Oriented Programming, Machine learning, Operating Systems, Computer Architecture and Organization, Web Design and Development, Human-Computer Interaction.

PROFESSIONAL EXPERIENCE

Deloitte Digital

Brisbane, Australia

Graduate Full-stack Software Developer

Jul 2023 - May 2024

- **Built the user authentication system by integrating Auth0 identity management** for Travel Money Oz, Australia's largest foreign exchange provider, securing access for **100,000+ users** transacting currency online.
- **Optimized customer authentication** by implementing Multi-Factor Authentication (MFA) and integrating mobile phone, passport, and ID verification, **reducing authentication-related support issues by ~70%** and enhancing security for same-day currency delivery.
- **Automated migration of 10,000+ user records** from a SQL Server database to Auth0 using C# encryption and decryption, completing the process in minutes and **reducing manual effort by 90%**.
- Spearheaded the creation of a **dashboard application** for employees using **REST APIs** with ASP.NET, and a **React.js (TypeScript)** front-end, enabling staff to monitor exchange rates and manage customer data with real-time SQL database updates.

Deloitte Digital

Brisbane, Australia

Summer Intern - Backend Software Developer

Nov 2021 - Dec 2021

- **Engineered a multi-currency payment feature** for Travel Money Oz using **ASP.NET and C#**, enabling users to add multiple currencies to travel cards seamlessly, improving **transaction efficiency by 25%** and boosting customer satisfaction.
- **Implemented backend APIs for real-time currency conversion and secure payment processing**, ensuring accuracy and scalability.

SKILLS

Python Libraries and Frameworks: scikit-learn, pandas, numpy, keras, PyTorch, pySerial, paho-mqtt.

Programming Languages: Python, C#, C++, Java, SQL, JSON, YAML files, JavaScript, Matlab, Objective-C, JavaScript

Frameworks and Libraries: .NET Framework, ASP.NET Web API, REST API, React.js, TypeScript, Node.js

Tools and Cloud Computing Platforms: Linux, GitHub, Postman, Jira, MySQL, Tableau, Bash, Docker, Jenkins, Kubernetes.

PUBLICATIONS

Visualising Physiological Cues in Individualised Virtual Reality for Mental Health

Oct 2023

- Presented at the **22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR)** under Dr. Nilufar Baghaei; conducted a literature review, defined design principles, and authored the research paper on individualized VR visualizations for mental health treatment.
- Independently **designed four distinct VR visualizations for physiological data**, created expert/user testing plans, and **conducted usability testing with 30+ participants**, refining designs for scalability and enhanced engagement.

SELECTED PROJECTS

AutoCareers – AI-Enhanced Job Search Platform

Feb 2025 - Present

- Created a **scalable full-stack AI job search platform** using React (TypeScript), FastAPI, MongoDB, and AWS, reducing candidate job-hunt time by **25%** through automated resume parsing, ATS scoring, and LLM-powered cover letter generation.
- **Led the development of 12+ responsive UI components** with Tailwind and Material UI, and integrated OpenAI/Together AI agents with asynchronous backend APIs, achieving **95% resume parsing accuracy**.
- Executed **end-to-end DevOps workflows**, including CI/CD pipelines on GitHub and AWS, API testing with Postman, and unit testing with Jest and Pytest, increasing system throughput by **30%** and ensuring production-grade stability.

Fairness-Aware Facial Recognition Models Using the Seldonian Framework

Sep 2024 - Dec 2024

- Devised a CNN-based fairness-aware **facial recognition model (PyTorch, scikit-learn)** using the Seldonian Framework **reducing racial bias by 20%**, achieving **95% compliance in age prediction tasks with fairness constraints**, and surpassing baseline models (ResNet50, VGG16).

Stock Price Prediction with Parallel Processing

Sep 2024 - Dec 2024

- Developed a **stock price prediction model using PySpark MLlib**, leveraging **parallel computing and SQL** to process large-scale financial data and **accelerated model training** by up to **65%** through multi-core partitioning.

Indoor Air Quality Indicator with Mechanical Plant Avatar

Feb 2023 - Jun 2023

- Directed **firmware development** of an embedded Indoor Air Quality system integrating environmental sensors and real-time IoT data streaming; implemented a traditional ML classifier achieving **80% accuracy in air-quality prediction**.